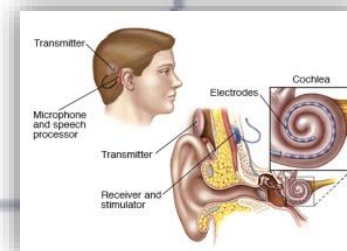
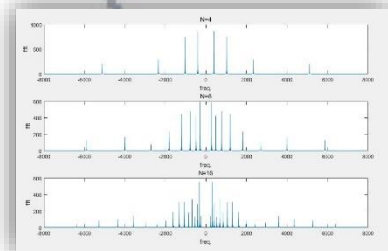
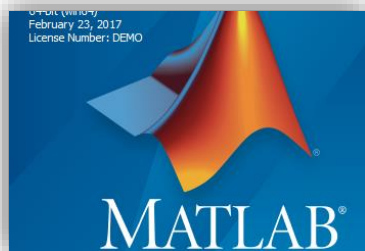


信号与系统实验

主讲人：吴光 博士

Email: wug@sustech.edu.cn



Signals and Systems (Lab)

Lab 4: The Continuous-Time Fourier Transform

Dr. **Wu Guang**

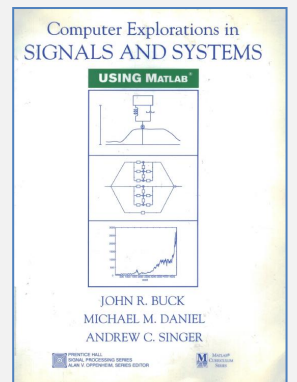
wug@sustech.edu.cn

**Electrical & Electronic Engineering
Southern University of Science and Technology**

Labs in this course—Review

5 Lab assignments+2 Projects

- Lab 1: MATLAB Programming (3 Weeks)
- Lab 2: Linear Time-Invariant Systems (2 Weeks)
- **Lab 3: Fourier Series Representation of Periodic Signals (2 Weeks)**
- Lab 4: The Continuous-Time Fourier Transform (2 Weeks)
- Lab 5: Coding Test & System, Transform, Convolution and Filter (1 Week)
- Project 1: Speech synthesis and perception with envelope cue (3 Weeks)
- Project 2: Motion detection via communication signals (3 Weeks)



Review—Complex Exponentials

$$\begin{aligned} x(t) = e^{st} &\longrightarrow \boxed{h(t)} \longrightarrow y(t) = \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau \\ &= \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau \right] e^{st} \\ &= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}} \end{aligned}$$

$s = j\omega$ - purely imaginary,
i.e. signals of the form $e^{j\omega t}$



The system H is LTI

Review—Exercise 1: Frequency response

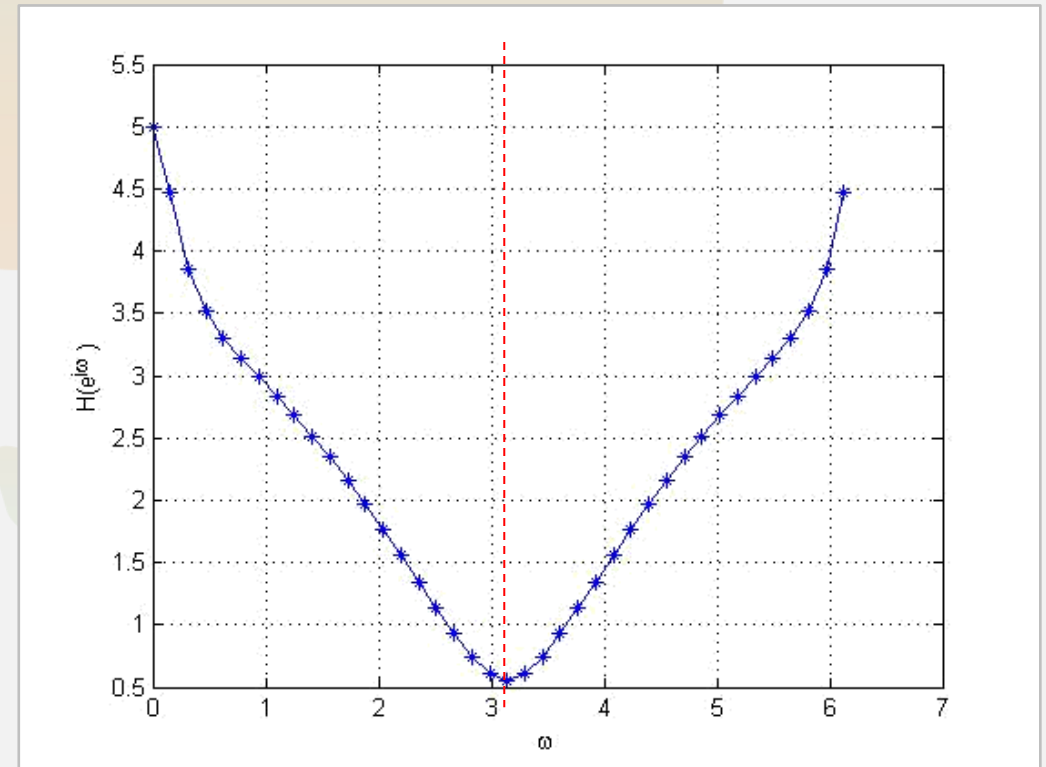
➤ Consider LTI System: $y[n] - 0.8y[n - 1] = 2x[n] - x[n - 2]$

Define the vector of coefficients:

```
A=[1 -0.8];  
B=[2 0 -1];
```

Plot the frequency response:

```
[H Omega] = freqz(B, A, 40, 'whole');  
plot(Omega, abs(H), '*-');  
xlabel('\omega');  
ylabel('H(e^{j\omega})');  
grid;
```



Review—Exercise 2: fft

```
A = fft(x) / length(x);  
figure(1)  
subplot(2,1,1),stem(abs(A));  
subplot(2,1,2),stem(abs(fftshift(A)));
```

```
>> fftshift([1 2 3 4 5 6])
```

```
ans =
```

```
4      5      6      1      2      3
```

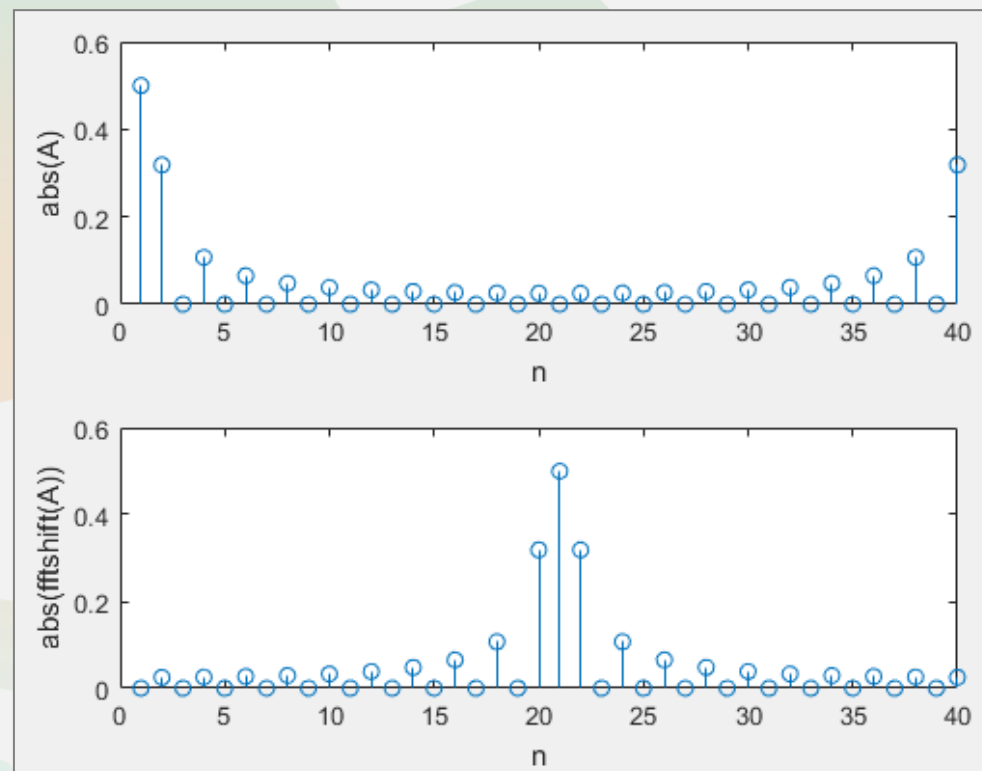
```
>> help fftshift
```

fftshift - 将零频分量移到频谱中心

此 MATLAB 函数 通过将零频分量移动到数组中心，重新排列傅里叶变换 x 。

```
Y = fftshift(X)
```

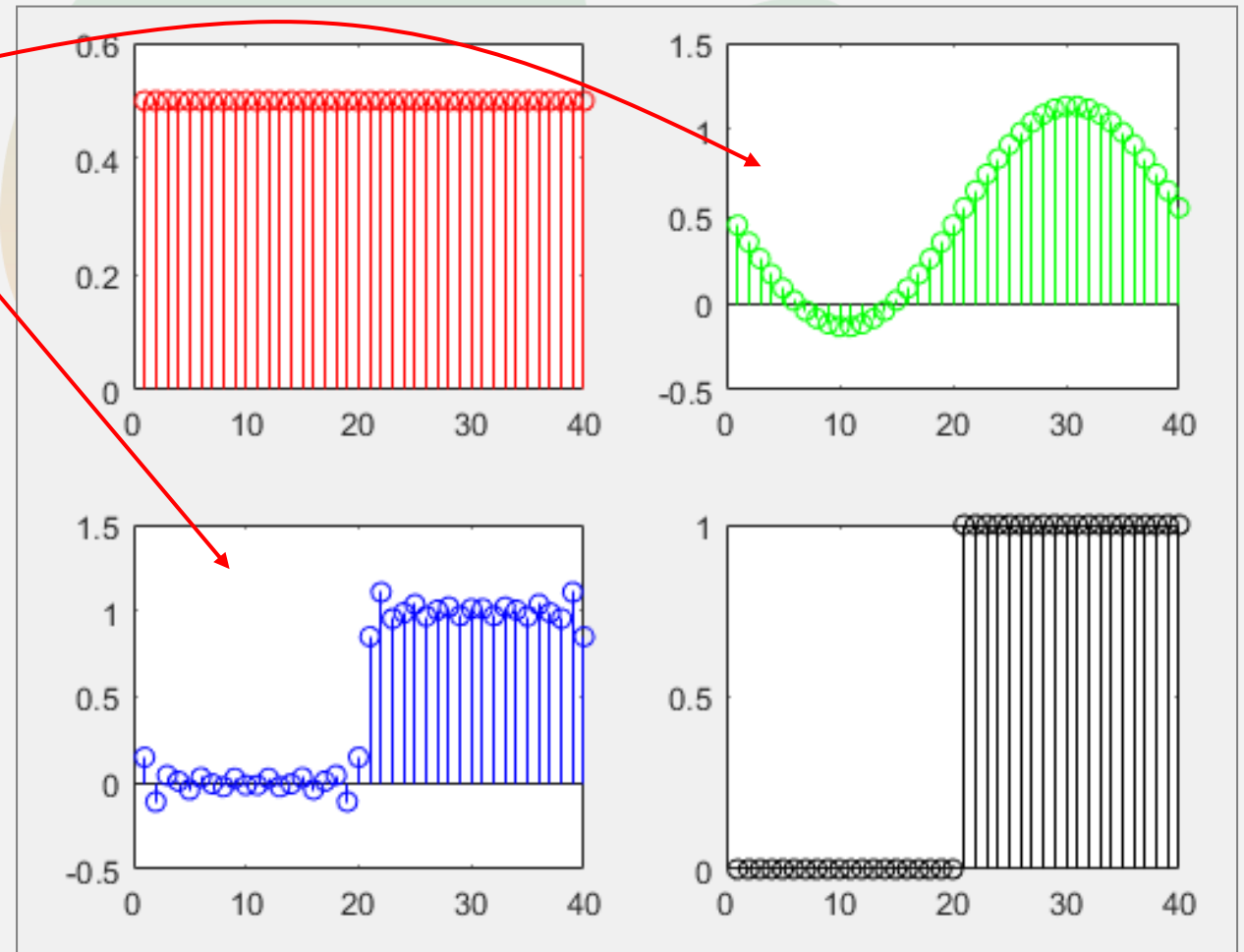
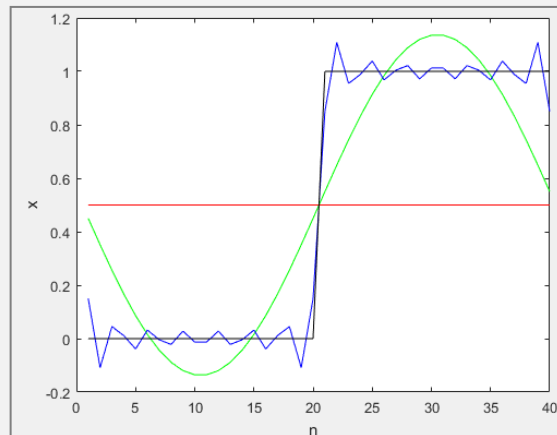
```
Y = fftshift(X,dim)
```



Review—Exercise 3: ifft

```
A1 = [A(1) zeros(1,39)];  
A2 = [A(1) A(2) zeros(1,37) A(40)];  
A3 = [A(1) A(2:15) zeros(1,11) A(27:40)];  
subplot(2,2,1), stem(1:40,ifft(A1)*40, 'r');  
subplot(2,2,2), stem(1:40,ifft(A2)*40, 'g');  
subplot(2,2,3), stem(1:40,ifft(A3)*40, 'b');  
subplot(2,2,4), stem(1:40,x, 'k');
```

Reconstruct signals



Review—DFT and FFT of a 1024*1024 image

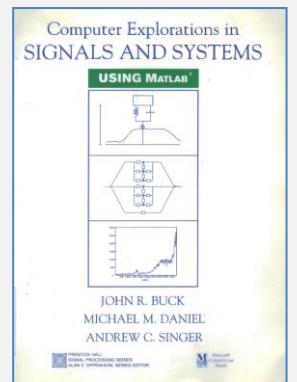
CPU	Clock Frequency	DFT	FFT
1941	60 Hz	152.3 y	271.4 d
1971 (4004)	108KHz	30.8 d	3.6 h
1978 (8086)	10MHz	8.0 h	2.3 min
1982 (80286)	20MHz	4.0 h	1.2min
1985 (80386)	33MHz	2.4h	42.6s
1989 (80486)	100MHz	48.0min	14.1s
1995 (Pentium)	200MHz	24.0min	7.0s
1999 (Pentium III)	450MHz	10.7min	3.1s
2000 (Pentium 4)	1.4GHz	3.4min	1.0s
2001 (Pentium 4)	2GHz	2.4min	0.7s

$O(N \log N)$

Labs in this course

5 Lab assignments+2 Projects

- Lab 1: MATLAB Programming (3 Weeks)
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- **Lab 4: The Continuous-Time Fourier Transform (2 Weeks)**
- Lab 5: Coding Test & System, Transform, Convolution and Filter (1 Week)
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- Project 2: Motion detection via communication signals (3 Weeks)



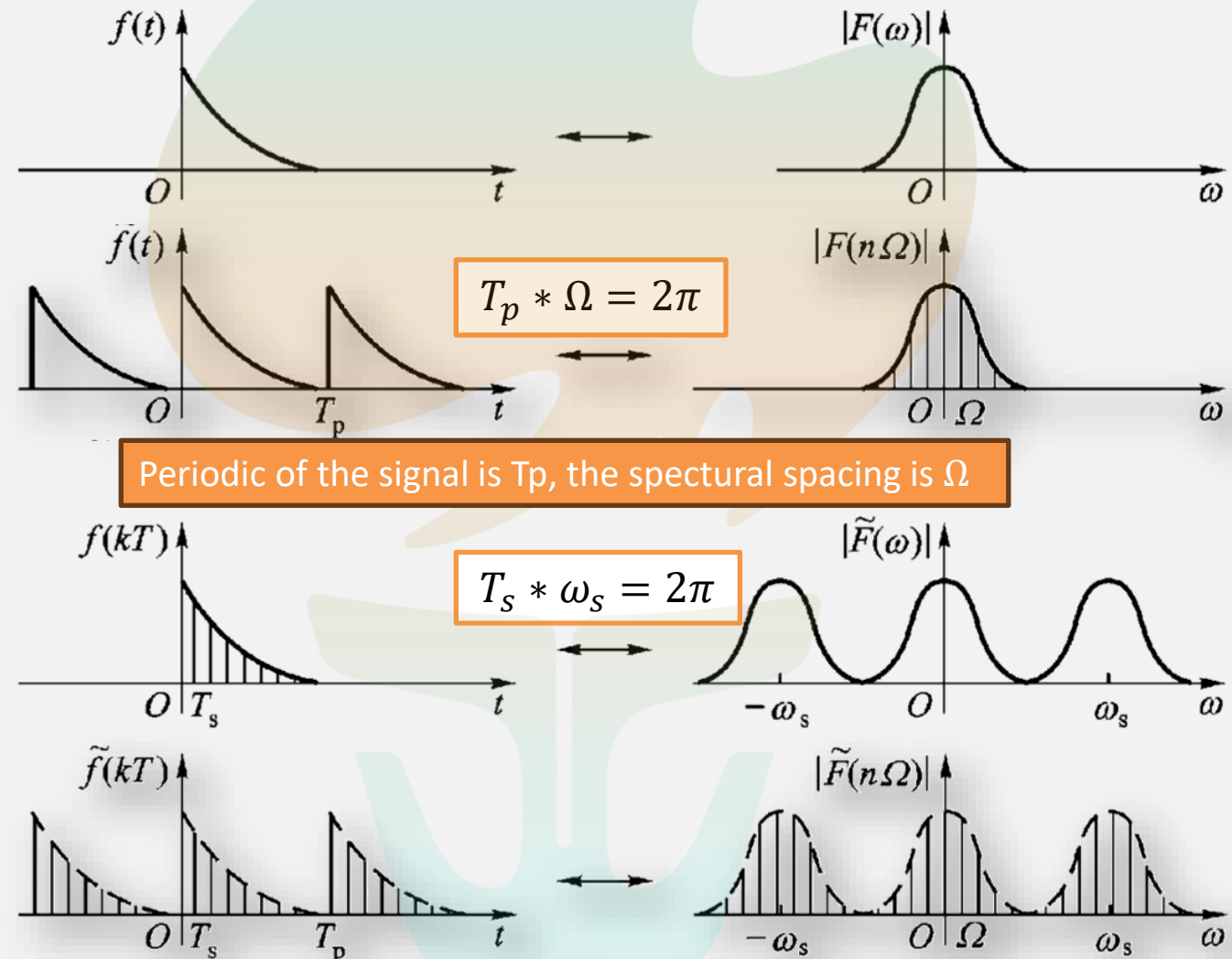
1. Periodic signal and aperiodic signal

FT
Fourier Transform

FS
Fourier Series

DFT
Discrete Fourier Transform

DTFS
Discrete Time Fourier Series



2. CT Fourier Transform

➤ Definition of CT Fourier transform:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

- Function of **continuous** frequency ω
- Represent the “**spectrum**” of a signal

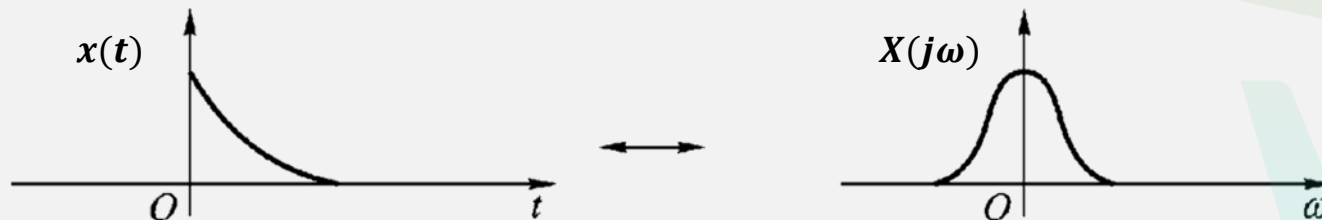
CT Fourier Transform:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$



DT Fourier Series:

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n]e^{-jk\left(\frac{2\pi}{N}\right)n}$$



3.1 Use `fft()` to calculate CTFT approximately

CT Fourier Transform:

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ &= \lim_{\tau \rightarrow 0} \tau \sum_{n=-\infty}^{\infty} x(n\tau) e^{-j\omega \tau n} \end{aligned}$$



DT Fourier Series:

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk \left(\frac{2\pi}{N} \right) n}$$

3.2 Use fft() to calculate CTFT approximately

- Suppose the dominant region of $x(t)$ is in $[0, T]$, then we can use the following approximation

$$\text{➤ } X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \approx \int_0^T x(t) e^{-j\omega t} dt \approx \sum_{n=0}^{N-1} x(n\tau) e^{-j\omega n\tau} \tau$$

- It could be observed that
- Slice into N intervals, each with length $\tau = T/N$

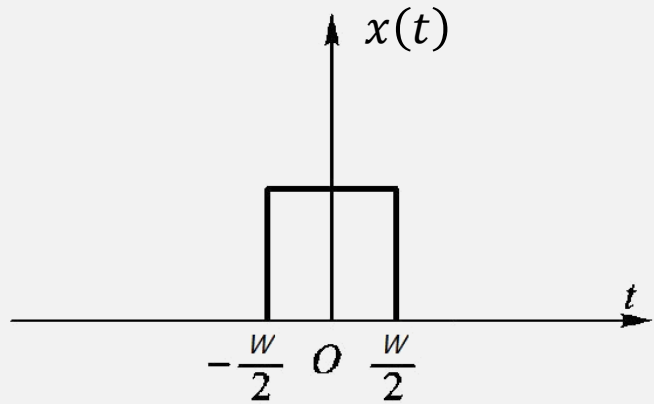
$$\text{➤ } X(j \frac{2\pi k}{N\tau}) \approx \tau \sum_{n=0}^{N-1} x(n\tau) e^{-j \frac{2\pi kn}{N}}$$

$$\text{➤ } k = \frac{1-N}{2}, \dots, -1, 0, 1, \dots, \frac{N-1}{2}$$

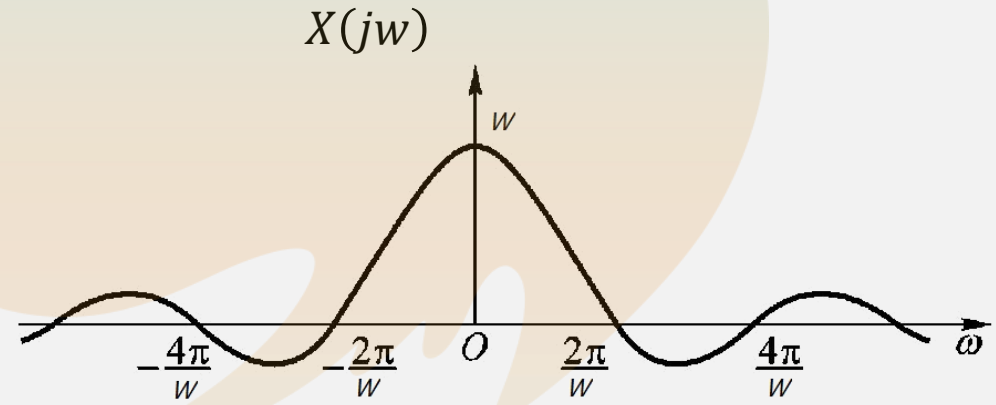
- **fft()** of $[x(0), x(\tau), x(2\tau), \dots, x((N-1)\tau)]$

- **Conclusion:** we can use **fft()** to calculate CTFT approximately, which could reduce the computation complexity

4.1 Example-sinc function



$$x(t) = \begin{cases} 1, & |t| < \frac{W}{2} \\ 0, & |t| > \frac{W}{2} \end{cases}$$



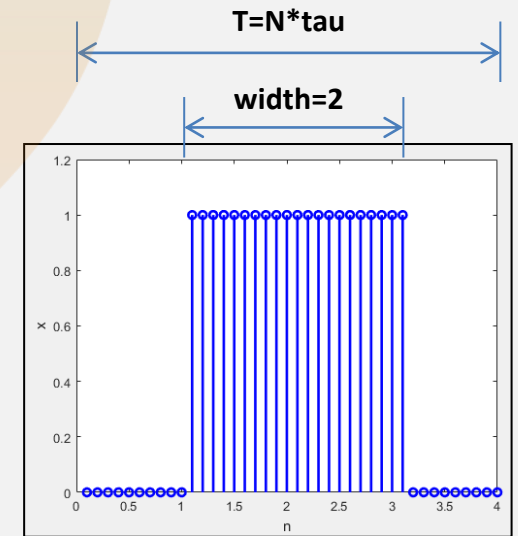
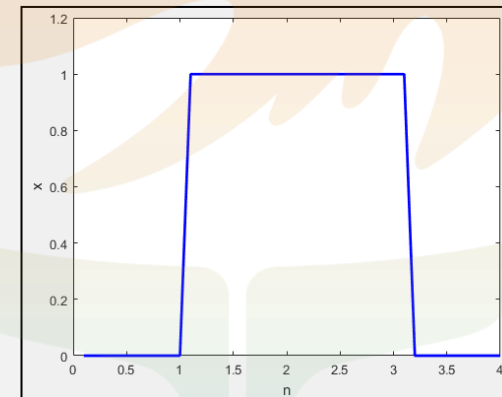
$$X(j\omega) = \left[\frac{\sin\left(\frac{\omega W}{2}\right)}{\frac{\omega W}{2}} \right] = W \text{Sinc}\left(\frac{\omega W}{2}\right)$$

4.2 Exercise 1

```
tau=0.5;
N=201; % N should be odd number, N*tau=T>3
x=zeros(1,1/tau),ones(1,2/tau+1),zeros(1,N-3/tau-1)];
figure(1)
plot([1:N]*tau,x,'r-','LineWidth',2)
xlabel('n');ylabel('x');
axis([0 4 0 1.2])
legend('\tau=0.5,N=201')
```

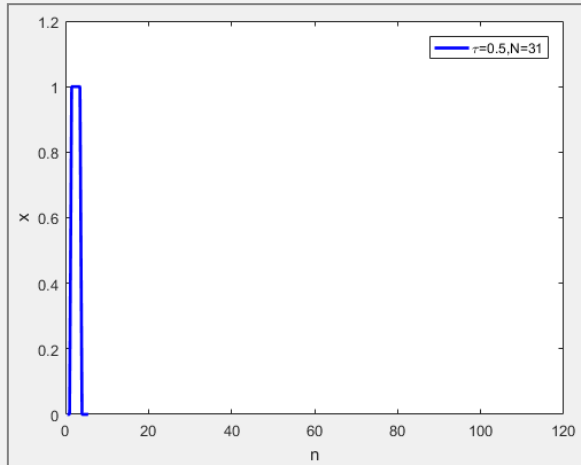
```
y=fftshift(fft(x));
X=tau*y;
lb = (1-N)*pi/N/tau;
ub = (N-1)*pi/N/tau;
step = 2*pi/N/tau;
figure(2)
plot(lb:step:ub, abs(X),'r-','LineWidth',2);
xlabel('\omega');ylabel('abs(X)');
legend('\tau=0.5,N=201')
```

- Single rectangular wave with width=2
- T should be larger than 3

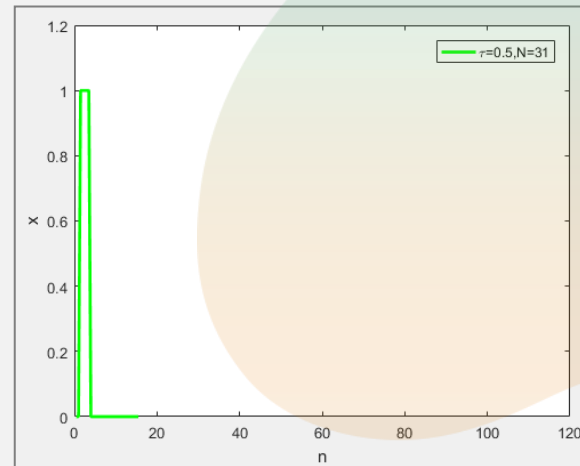


What happen if we change tau and N? Why?

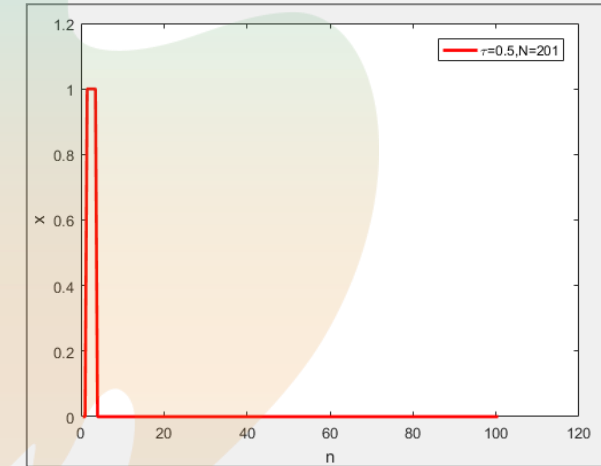
4.3 Results—Increase N



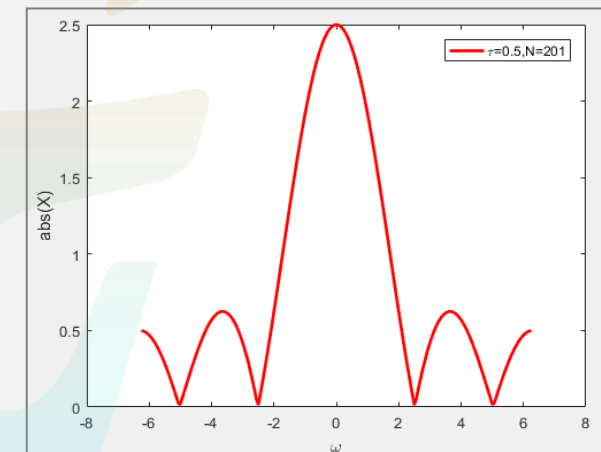
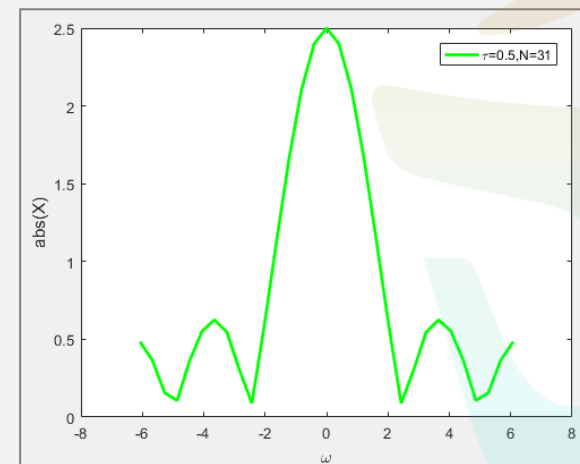
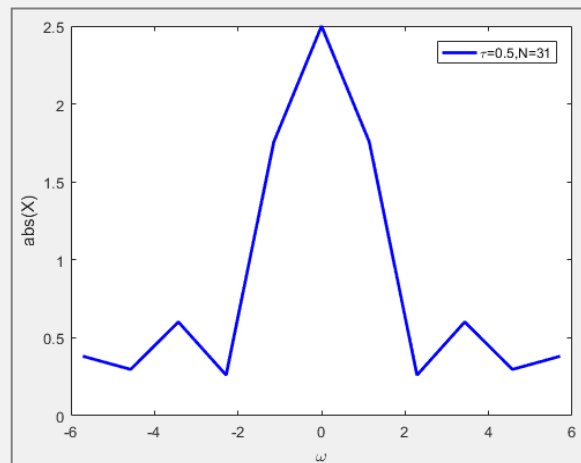
N=11



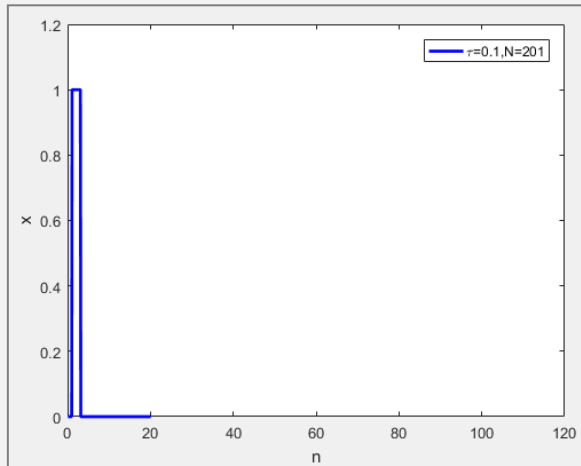
N=31



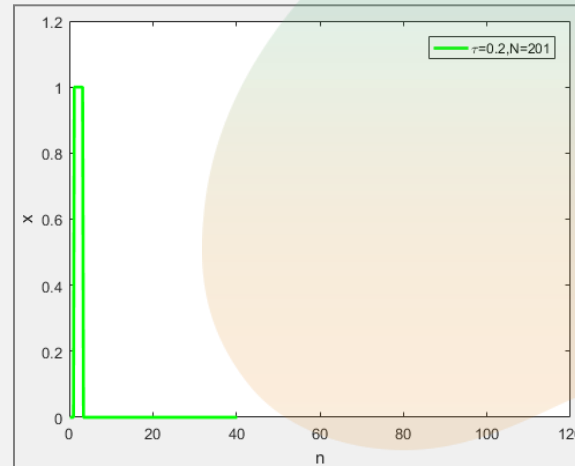
N=201



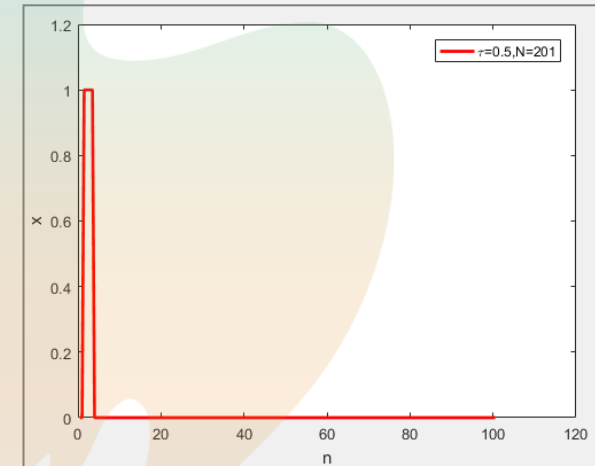
4.4 Results—Decrease τ



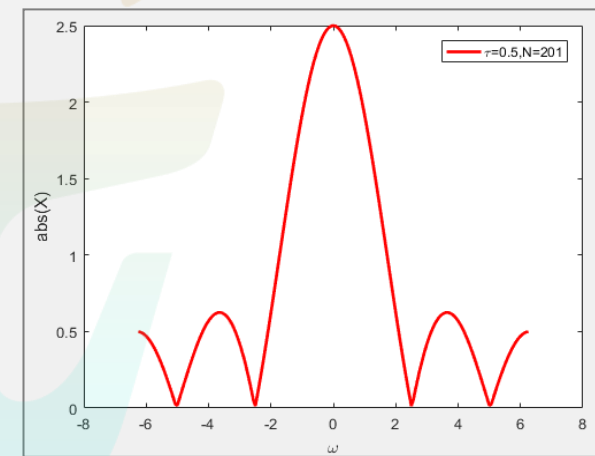
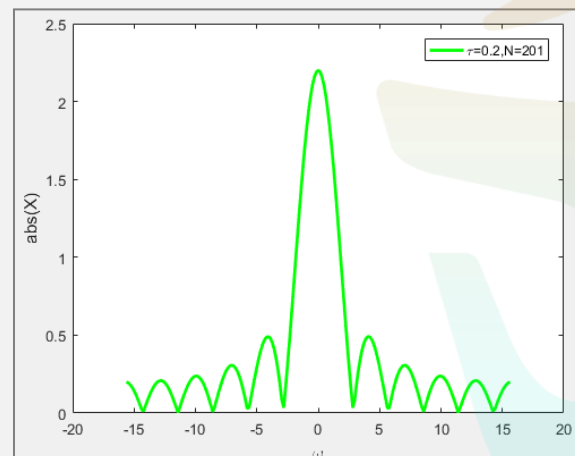
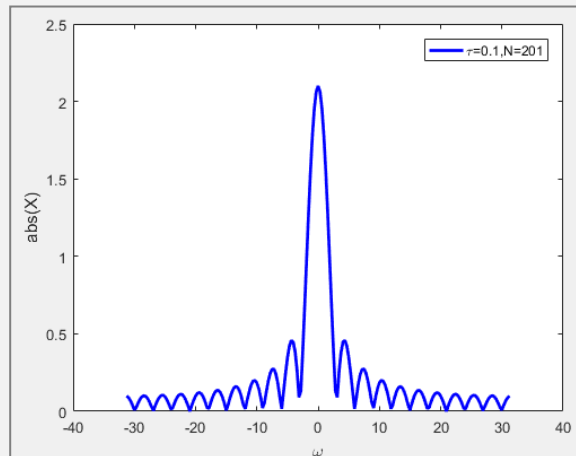
$\tau = 0.1$



$\tau = 0.2$



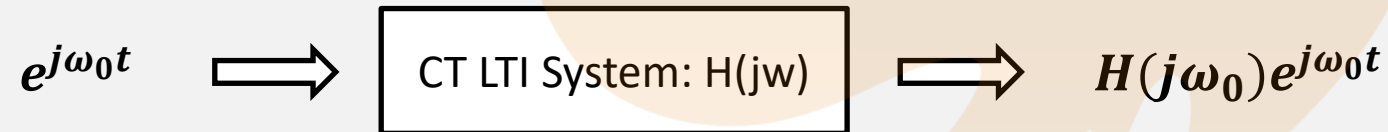
$\tau = 0.5$



5.1 What is Frequency Response ?

➤ Frequency response

- A function of frequency
- System gain in frequency domain

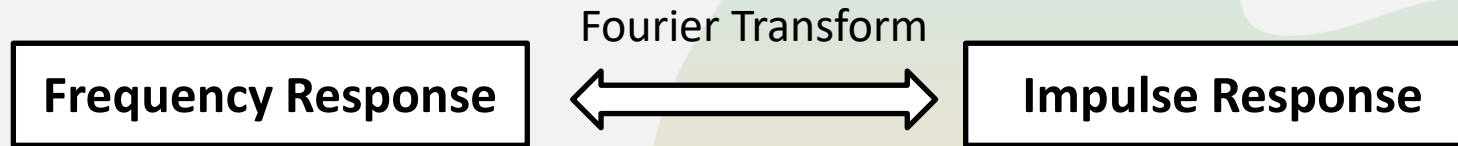


➤ Math definition

- Fourier transform of impulse response

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t} dt = Y(j\omega)/X(j\omega)$$

5.2 Frequency Response and Impulse Response



➤ Transform pair:

$$\text{➤ } e^{-at}u(t) \quad \text{Re}\{a\} > 0 \quad \Leftrightarrow \quad \frac{1}{j\omega + a}$$

➤ Example:

$$\text{➤ } \frac{3}{j\omega + 3} \Leftrightarrow 3e^{-3t}u(t)$$

➤ How about general fraction?

$$H(j\omega) = \frac{b_M(j\omega)^M + b_{M-1}(j\omega)^{M-1} + \dots + b_1(j\omega) + b_0}{a_N(j\omega)^N + a_{N-1}(j\omega)^{N-1} + \dots + a_1(j\omega) + a_0}$$

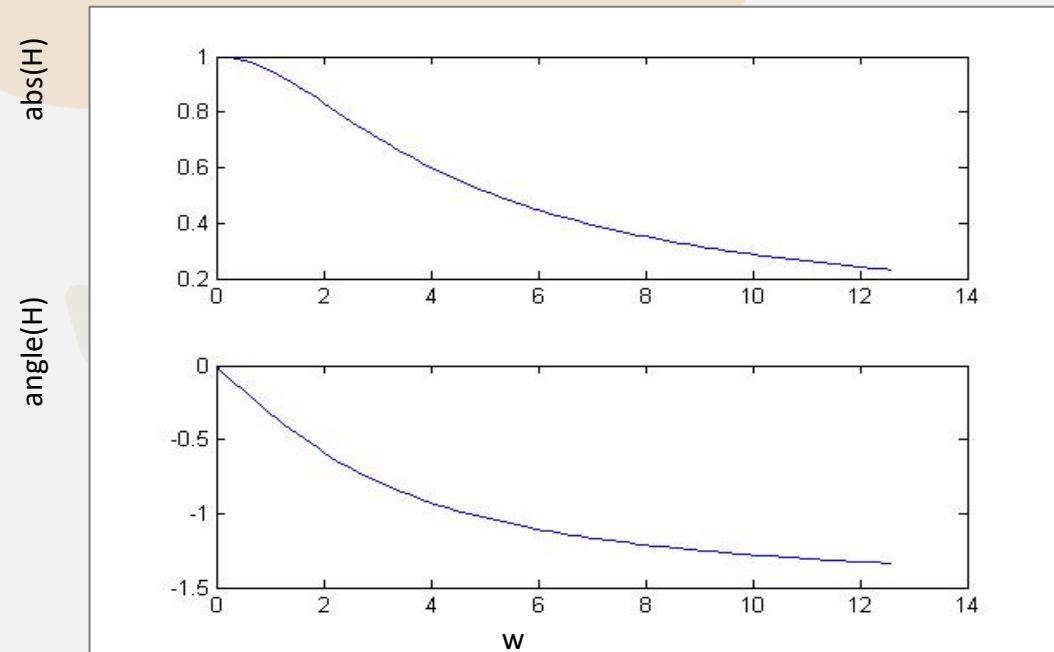
5.3 Function: freqs()

- **Description:** generate the frequency response of LTI system
- **Syntax:** freqs(b,a,w)
- **Example:**

$$\frac{dy(t)}{dt} + 3y(t) = 3x(t)$$

```
a=[1 3];  
b=3;  
w=linspace(0,4*pi);  
H=freqs(b,a,w);  
subplot(2,1,1), plot(w,abs(H));  
subplot(2,1,2),  
plot(w,angle(H));
```

High-pass or Low-pass?



6.1 Differential Equation and Frequency Response

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

$$H(j\omega) = \frac{b_M(j\omega)^M + b_{M-1}(j\omega)^{M-1} + \cdots + b_1(j\omega) + b_0}{a_N(j\omega)^N + a_{N-1}(j\omega)^{N-1} + \cdots + a_1(j\omega) + a_0}$$

Proof:

$$\begin{aligned} \mathcal{F} \left[\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} \right] &= \sum_{k=0}^N \mathcal{F} \left[a_k \frac{d^k y(t)}{dt^k} \right] = \sum_{k=0}^N a_k (j\omega)^k Y(j\omega) \\ \mathcal{F} \left[\sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k} \right] &= \sum_{k=0}^M \mathcal{F} \left[b_k \frac{d^k x(t)}{dt^k} \right] = \sum_{k=0}^M b_k (j\omega)^k X(j\omega) \end{aligned}$$

6.2 Partial Fraction Expansion

➤ Partial Fraction Expansion (No identical poles):

$$H(j\omega) = \frac{b_M(j\omega + z_1)(j\omega + z_2) \dots (j\omega + z_M)}{a_N(j\omega + p_1)(j\omega + p_2) \dots (j\omega + p_N)} = \frac{A_1}{j\omega + p_1} + \frac{A_2}{j\omega + p_2} + \dots + \frac{A_N}{j\omega + p_N}$$

➤ Partial Fraction Expansion (with identical poles):

$$H(j\omega) = \frac{b_M(j\omega + z_1)(j\omega + z_2) \dots (j\omega + z_M)}{a_N(j\omega + p_1)^{k_1}(j\omega + p_2)^{k_2} \dots (j\omega + p_n)^{k_n}} = \frac{A_{1,1}}{(j\omega + p_1)^{k_1}} + \frac{A_{1,2}}{(j\omega + p_1)^{k_1-1}} + \dots + \frac{A_{1,k_1}}{(j\omega + p_1)} + \dots + \frac{A_{n,1}}{(j\omega + p_n)^{k_n}} + \frac{A_{n,2}}{(j\omega + p_n)^{k_n-1}} + \dots + \frac{A_{n,k_n}}{(j\omega + p_n)}$$

6.3 Partial Fraction Expansion

Therefore, we should know the FT of

$$\frac{1}{(j\omega + a)^k}$$

We already know:

$$e^{-at}u(t) \quad \text{Re}\{a\} > 0 \quad \Leftrightarrow \quad \frac{1}{j\omega + a}$$

According to the FT property:

$$tx(t) \Leftrightarrow j \frac{d}{d\omega} X(j\omega)$$

We have:

$$te^{-at}u(t) \quad \text{Re}\{a\} > 0 \quad \Leftrightarrow \quad \frac{1}{(j\omega + a)^2}$$

...

6.4 Function: residue()

- Convert between partial fraction expansion and polynomial coefficients
- $[r,p,k] = \text{residue}(b,a)$; $[r,p] = \text{residue}(b,a)$

$$\frac{b(s)}{a(s)} = \frac{b_1 s^m + b_2 s^{m-1} + b_3 s^{m-2} + \dots + b_{m+1}}{a_1 s^n + a_2 s^{n-1} + a_3 s^{n-2} + \dots + a_{n+1}}$$

$$\frac{b(s)}{a(s)} = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \dots + \frac{r_n}{s - p_n} + k(s)$$

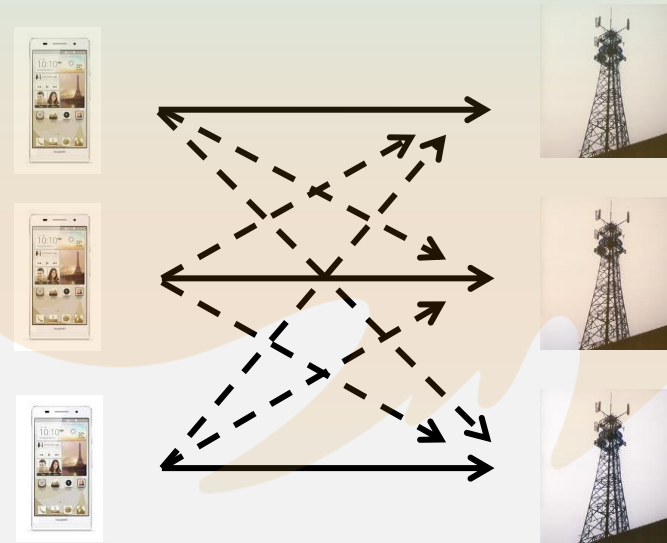
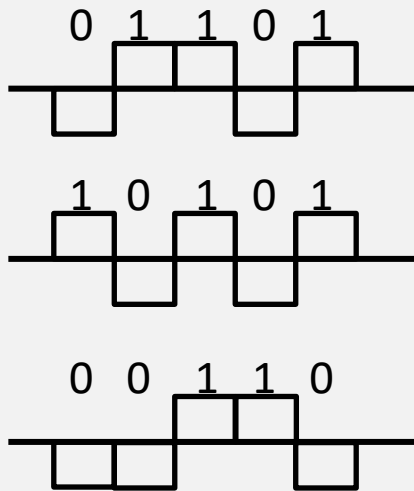
- Example:

$$\frac{b(s)}{a(s)} = \frac{5s^3 + 3s^2 - 2s + 7}{-4s^3 + 8s + 3}$$



```
b = [ 5 3 -2 7];  
a = [-4 0 8 3];  
[r, p, k] = residue(b,a);
```


7.1 Application: What is Modulation ?



- Each phone wants to deliver information to its base station
- Therefore, there is cross-talk in the wireless channel
- **How can we solve this issue?**

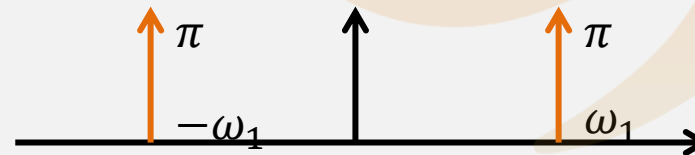
7.2 Application: Amplitude Modulation

- Signal: $m(t)$



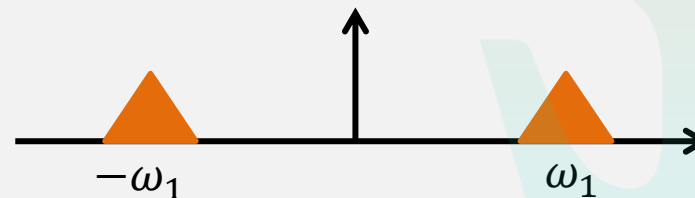
Spectrum of $m(t)$

- Carrier: $\cos(\omega_1 t) \Leftrightarrow \pi[\delta(\omega - \omega_1) + \delta(\omega + \omega_1)]$



Spectrum of carrier

- Amplitude Modulation: $y(t) = m(t) \times \cos(\omega_1 t)$



Shift by ω_1 and Scale by $1/2$

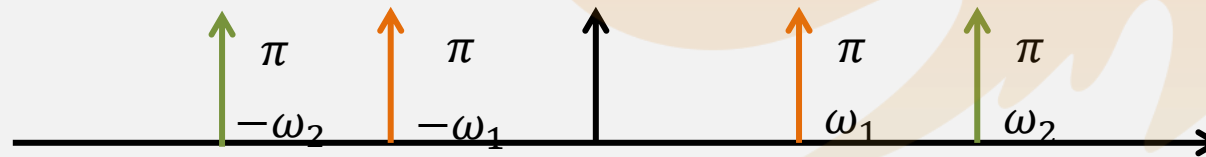
No information loss

7.3 Application: Amplitude Modulation

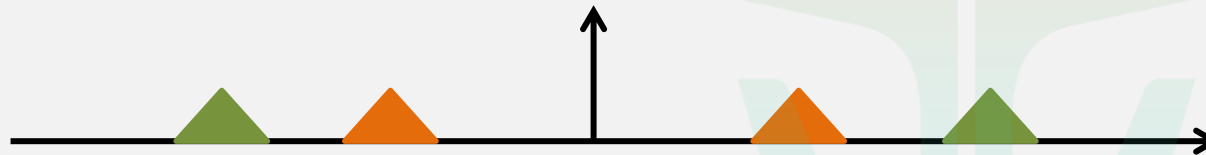
- Signals: $m_1(t)$ and $m_2(t)$



- Carriers: $\cos(\omega_1 t)$ and $\cos(\omega_2 t)$



- Amplitude Modulation: $y(t) = m_1(t) \times \cos(\omega_1 t) + m_2(t) \times \cos(\omega_2 t)$



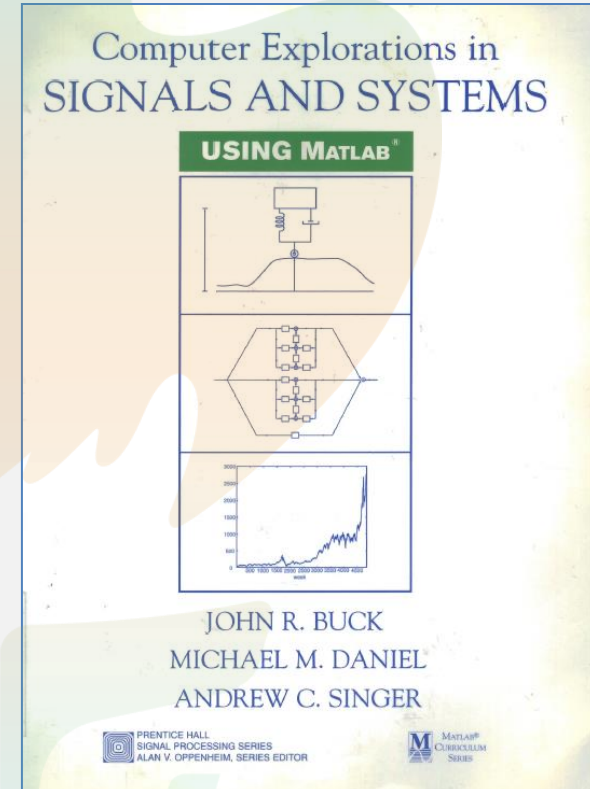
Signals are distinguished in frequency domain !

Lab Assignments

- Read tutorial 4.1 by yourself
- 4.2 & 4.6

$$x(t) = m_1(t) \cos(2\pi f_1 t) + m_2(t) \sin(2\pi f_2 t) + m_3(t) \sin(2\pi f_1 t)$$

- Submit your report.



Question ?

